

Anshul Modi

Artificial Intelligence Engineer

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TECHNICAL SKILLS

Programming: Python, C, C++, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Transformers, Large Language Models

AI Systems: RAG, Vector Databases, Agentic AI, Multi-Agent Orchestration

Frameworks: PyTorch, Scikit-learn, LangChain, LangGraph, PyTorch Geometric

Tools: Git, FastAPI, Pathway, SHAP, XGBoost, Pandas, NumPy

Core: Data Structures & Algorithms, Real-time Data Pipelines, Graph Neural Networks

EDUCATION

Indian Institute of Technology (IIT), Dharwad

2024 – 2028

B.Tech in Chemical Engineering

Class XII: 92% — Class X: 95%

KEY PROJECTS

Multimodal Fraud Detection System | Python, PyTorch, XGBoost, BERT, GraphSAGE, PyTorch Geometric, SHAP | GitHub

- Built end-to-end multimodal pipeline combining tabular (XGBoost), NLP (fine-tuned BERT), and graph (GraphSAGE) modalities on 590k transactions, achieving AUC-ROC 0.9502 and PR-AUC 0.7055 on a 28:1 imbalanced dataset
- Designed late-fusion attention MLP dynamically weighting each modality; graph topology contributed 43% attention weight, nearly matching raw tabular features in predictive power
- Implemented GraphSAGE on 13,944-node transaction network using PyTorch Geometric; ablation confirmed +6.2% AUC gain over best single-modality baseline (XGBoost at 0.8885)
- Applied focal loss ($\alpha=0.25$, $\gamma=2$) and Platt scaling; tuned decision threshold (0.38) on validation set, achieving F1 score of 0.6613

Agentic AI HR Automation System | Python, FastAPI, LangChain, LangGraph, RAG | GitHub

- Designed and implemented a multi-agent AI system for automated recruitment workflows, reducing manual screening effort by ~70%
- Built modular agents for resume parsing, job description generation, and candidate evaluation using LLMs
- Integrated RAG pipeline with vector databases for semantic job–resume matching, achieving ~85% top-3 match accuracy
- Developed async FastAPI backend APIs with ~120ms median response time for real-time processing and system orchestration

YouTube Sentiment Analysis & Browser Extension | Python, NLP, Scikit-learn, JavaScript | GitHub

- Built end-to-end NLP pipeline for sentiment classification of YouTube comments, reaching 89% accuracy
- Improved model accuracy from 74% to 89% using preprocessing, feature engineering, and hyperparameter tuning
- Developed browser extension for real-time sentiment visualization on YouTube
- Engineered live comment-stream processing delivering insights in under 2s

ChemSafety Copilot | Python, FastAPI, Streamlit, Groq LLM API, ChromaDB, RAGAS | GitHub

- Built an agentic RAG pipeline over 20 CSB reports with a 5-intent router (RAG, PubChem, PSV sizing, comparative)
- Engineered hybrid dense + BM25 + RRF retrieval with cross-encoder reranking and a CRAG retry loop, lifting Recall@5 from 0.682 to 0.818 and MRR from 0.511 to 0.674
- Implemented independent LLM faithfulness verification and HyDE/multi-query/step-back query expansion
- Added an auto-generated SVG diagram pipeline (PSV, bowtie, GHS pictograms) and a 29-test pytest suite

HONORS & ACHIEVEMENTS

- Secured 4th Position – Nasiko AI Hackathon (HR AI Management) among competitive student teams
- Achieved Distinction in NTSE (National Talent Search Examination), reflecting a strong academic foundation
- Solved 100+ problems on LeetCode, focusing on Data Structures & Algorithms and competitive coding

LEADERSHIP & EXTRACURRICULARS

- **Inter IIT Cultural Meet – Quiz Cup Representative:** Represented IIT Dharwad at the Inter IIT Cultural Meet, competing against students from premier IITs across India
- **Member, Coding Club – IIT Dharwad:** Actively participate in hackathons, coding contests, and collaborative AI/ML projects; contributed to club-level workshops on Python and machine learning
- **Member, Quiz Club – IIT Dharwad:** Engage in inter-college quizzes and internal competitions, fostering analytical thinking, lateral reasoning, and broad general knowledge

AREAS OF INTEREST

Artificial Intelligence • Natural Language Processing • LLM Systems • Agentic AI • Graph Neural Networks • Multimodal ML